

UNIT-4 | Lecture 1

Domain & Data Dependency + Functional Dependencies

Coverage: Hour 1

Learning Objectives

Understand domains and data dependencies.

Understand functional dependency (FD), partial dependency, and transitive dependency.

Domain Dependency

A **domain** is the set of valid values for an attribute. It defines properties such as data type, format, range, and allowed values.

Example: Age may use an integer domain; Branch may use a defined set of valid branch names.

Data Dependency

Data dependency describes how attributes are related. The key dependency used in normalization is **functional dependency**.

$X \rightarrow Y$ means that if two tuples agree on X, they must agree on Y. X is the determinant.

Examples

Student_ID $\rightarrow$ Name, Branch: Student_ID determines the student's name and branch.

Partial dependency: a non-prime attribute depends on part of a composite candidate key.

Transitive dependency: $X \rightarrow Y$ and $Y \rightarrow Z$ imply $X \rightarrow Z$, where the dependency through Y violates the desired normal-form condition.

Importance

Dependencies help identify candidate keys, redundancy, update anomalies, and normalization requirements.

They provide the basis for decomposition into well-structured relations.

Key Takeaway

Functional dependencies describe attribute relationships and are the foundation of normalization.